Machine Learning: Unsupervised Learning

Part – 2

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Principal Component Analysis

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Content

- What Is Principal Component Analysis?
- How PCA Constructs the Principal Components
- Step-by-Step Explanation of PCA

What Is Principal Component Analysis?

- Principal component analysis, or PCA, is a dimensionality reduction method that is often used to reduce the dimensionality of large data sets, by transforming a large set of variables into a smaller one that still contains most of the information in the large set.

What Is Principal Component Analysis?

- Reducing the number of variables of a data set naturally comes at the expense of accuracy, but the trick in dimensionality reduction is to trade a little accuracy for simplicity.

- Because smaller data sets are easier to explore and visualize, and thus make analyzing data points much easier and faster for machine learning algorithms without extraneous variables to process.

- So, to sum up, the idea of PCA is simple: reduce the number of variables of a data set, while preserving as much information as possible.

What Is Principal Component Analysis?

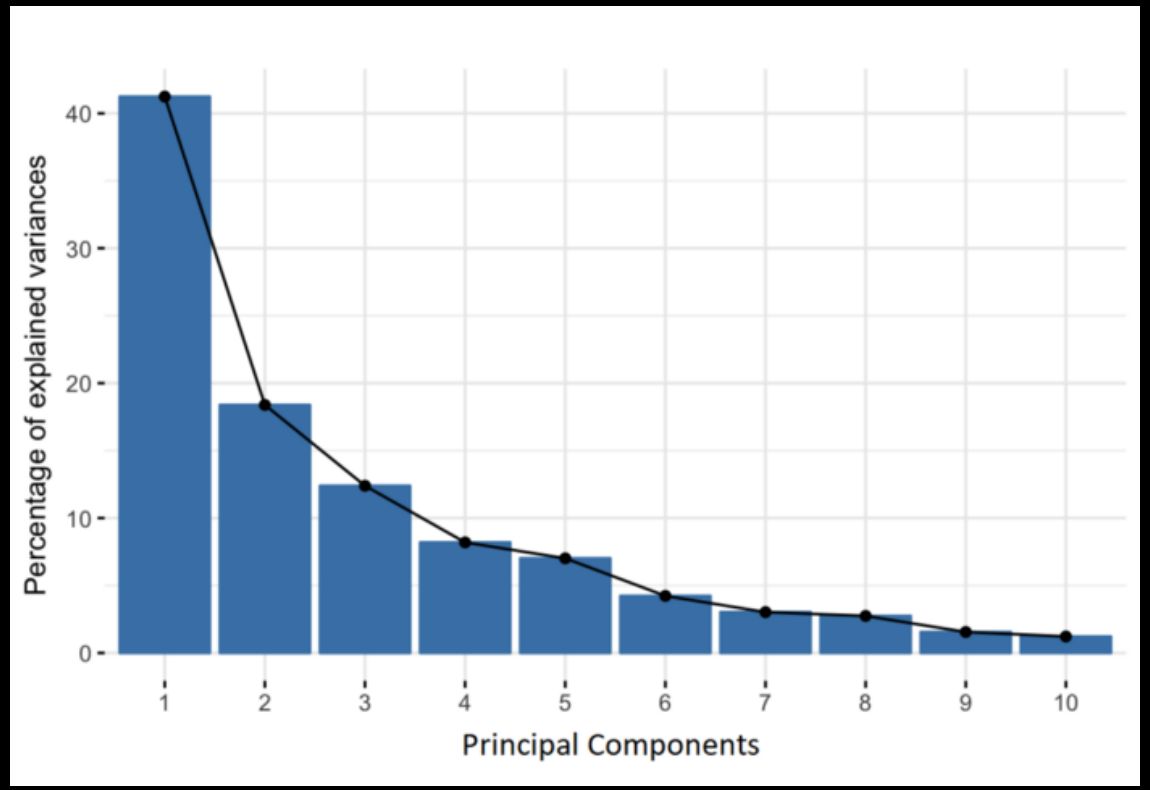
- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables. These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.

- So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on.

What Is Principal Component Analysis?

- Organizing information in principal components this way will allow you to reduce dimensionality without losing much information, and this by discarding the components with low information and considering the remaining components as your new variables.

- An important thing to realize here is that the principal components are less interpretable and don't have any real meaning since they are constructed as linear combinations of the initial variables.



What Is Principal Component Analysis?

- Geometrically speaking, principal components represent the directions of the data that explain a maximal amount of variance, that is to say, the lines that capture most information of the data.

- The relationship between variance and information here, is that, the larger the variance carried by a line, the larger the dispersion of the data points along it, and the larger the dispersion along a line, the more information it has.

- To put all this simply, just think of principal components as new axes that provide the best angle to see and evaluate the data, so that the differences between the observations are better visible.

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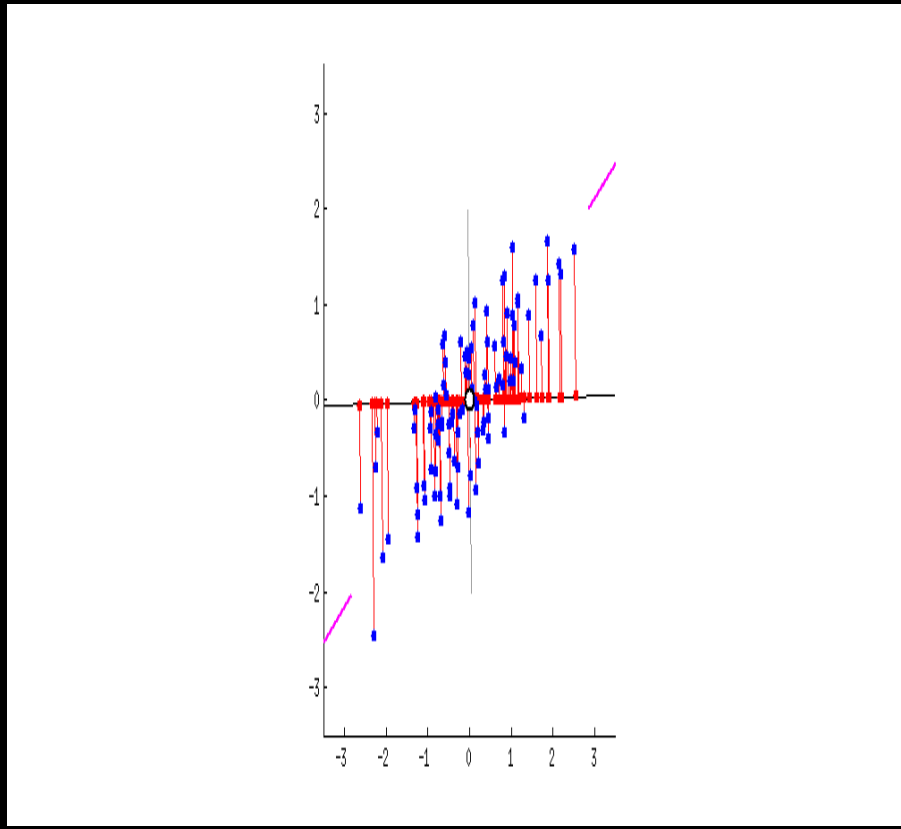
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How PCA Constructs the Principal Components

- As there are as many principal components as there are variables in the data, principal components are constructed in such a manner that the first principal component accounts for the largest possible variance in the data set.

How PCA Constructs the Principal Components

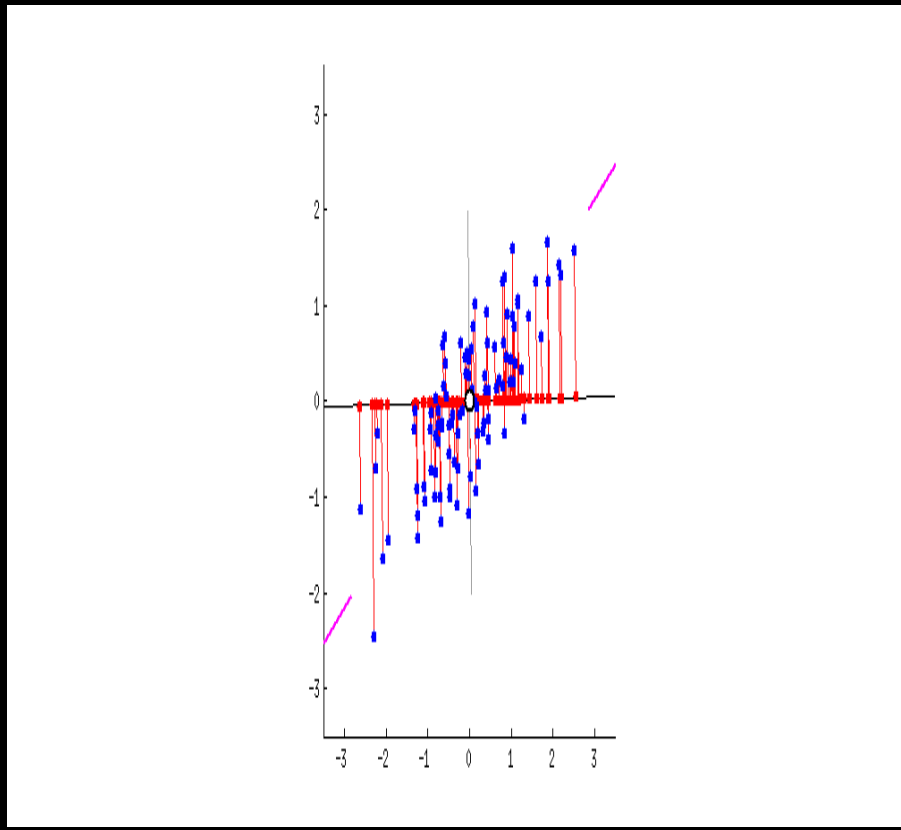
- For example, let's assume that the scatter plot of our data set is as shown below, can we guess the first principal component ?
- Yes, it's approximately the line that matches the purple marks because it goes through the origin and it's the line in which the projection of the points (red dots) is the most spread out.
- Or mathematically speaking, it's the line that maximizes the variance (the average of the squared distances from the projected points (red dots) to the origin).



How PCA Constructs the Principal Components

- The second principal component is calculated in the same way, with the condition that it is uncorrelated with (i.e., perpendicular to) the first principal component and that it accounts for the next highest variance.

- This continues until a total of *p* principal components have been calculated, equal to the original number of variables.



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Step-by-Step Explanation of PCA

➢ Step 1: Calculate mean:

$$\bar{X} = \frac{\sum_{i=1}^{m} x_i}{m}$$

➢ Step 2: Calculation of covariance matrix:

$$S = \begin{bmatrix} Cov(X_1, X_1) & Cov(X_1, X_2) \\ Cov(X_2, X_1) & Cov(X_2, X_2) \end{bmatrix}$$

$$Cov(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Step-by-Step Explanation of PCA

➢ Step 3: Calculate eigenvalues of the covariance matrix:

$$0 = \det(S - \lambda I)$$

where, I *is* $n \times n$ *identity matrix* .

➢ Step 4: Calculation of eigenvectors:

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda I)U$$

Step-by-Step Explanation of PCA

➢ Step 5: Computation of first principal components:

$$Transformed\ Data = X^{T}_{centered} \bullet u_1$$

Step-by-Step Explanation of PCA

➢ Find the first principal components of the dataset given below.

4	11
8	4
13	5
7	14



-4.3052
3.7361
5.6928
-5.1238




THANK YOU